Title

Methods for including environmental impacts in health economic evaluations and health technology assessments: a scoping review

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Introduction

Human health is intrinsically linked with planetary health and depends on flourishing natural systems and the wise stewardship of those systems to thrive.(1) But planetary resources are being degraded at an unprecedented rate and this poses an existential threat to human health and the sustainably of our healthcare systems.

Evidence from around the world suggests that health systems are significant contributors to the total greenhouse gas emissions (GHG) of a country. Studies have found that national healthcare systems are responsible for 7% of GHG in Australia,(2) 4% in England,(3) and 10% in the United States.(4) Healthcare professionals have responded to this challenge by embracing "sustainable healthcare" practices,(5,6) to minimise their impact on anthropogenic climate change, planetary health, and human health. This includes calls for the preferential use of technologies with lower environmental impact.(5)

Health technology assessment, and the health economic evaluations they rely on, may consider the environmental impact of a technology, and help decision makers choose low carbon technologies for reimbursement.(7–9) There are some published examples in the literature that address how environmental impacts could be included in these methods, however there is not yet any consensus or agreed approach for how this should be done.(10,11)

We hypothesise that environmental impacts could be integrated into health technology assessment in one of the following four ways:

1. Use environmental impacts to modify a decision based on a traditional health economic evaluation during health technology assessment
2. Include environmental impacts as one criterion of a multi criteria decision analysis (MCDA)
3. Convert environmental impacts into a monetary value and include them in the costs and/or benefits of a cost-benefit analysis (CBA)
4. Include environmental impacts as an additional societal cost in a cost-effectiveness analysis (CEA) or cost-utility analysis (CUA)

The objective of this review was to collate published descriptions of methods for including environmental impacts in health economic evaluations and health technology assessments. We identified a scoping review by Desterbecq and colleagues (12,13) and used it as a starting point for an updated review.

Review Questions

The aim of this scoping review was to determine the evidence that exists about methods to include environmental impacts in health economic evaluations and health technology assessments (HTA). We will consider what methods for including environmental impacts in HTA and health economic evaluations have been taken, explored, or recommended.

Keywords

Health technology assessment, environmental impact, economic evaluation, scoping review, health economics.

Eligibility Criteria

Participants, Concept, Context (PCC)

Sources that meet the following "PCC" (participants, concept, context) criteria will be considered for inclusion in the scoping review.

Participants	All human participant types and health conditions will be considered.
Concept	Methods, tools, guidance, examples, commentary, and all other information related to the inclusion of environmental impacts in health economic evaluations and health technology assessments.
Context	All contexts will be considered.

Types of Sources

This scoping review will consider both experimental and non-experimental study designs, including economic evaluations conducted alongside randomised controlled trials, models, non-randomised trials, and other study types. Qualitative studies, reviews, and opinion papers will also be considered for inclusion, as well as government and organisational publications, and other grey literature.

Methods

The proposed scoping review will be conducted in accordance with the JBI methodology for scoping reviews and reported according to the Preferred Reporting Items for Systematic Reviews and Meta Analyses Extension for Scoping Reviews (PRISMA-ScR) (14)

Search strategy

The search strategy will aim to locate both published and unpublished studies. An initial limited search of MEDLINE and Scopus was undertaken to identify articles on the topic.(7–10,15–20) The text words contained in the titles and abstracts of relevant articles, and the index terms used to describe the articles were used to develop a full search strategy in consultation with an academic librarian. Databases that will be searched are: MEDLINE, Scopus, EconLit, Environment Complete, and Evidence-Based Medicine Reviews (see Appendix 1 for Medline and Environment Complete search strategies). The authors of the pre-identified scoping review (12,13) searched PubMed, Embase, and Scopus between 1 June and 15 July 2021 and this search was considered comprehensive. We therefore restricted our search of MEDLINE and Scopus to results published during or after 2021. EconLit, Environment Complete, and Evidence-Based Medicine Reviews will be searched with no date restrictions. The 43 studies included by Desterbecq and colleagues will be screened for inclusion in this review. In addition, grey literature will be searched using 'Grey Matters: A Practical Tool for Searching Health-Related Grey Literature' and Google. The reference lists of included studies will be screened, as well as forward citations identified using Scopus and Google Scholar. Studies in any language will be considered for inclusion and Google Translate will be used when needed.

Source of Evidence Selection

Following the search, all identified sources of information will be collated using EndNote X9 (Clarivate Analytics, PA, USA) and uploaded into Covidence (www.covidence.org). Duplicates will be removed. Titles and abstracts will then be screened by JTWWW for assessment against the inclusion criteria for the review. Potentially relevant sources will be retrieved in full, and the full text will be assessed in detail against the inclusion criteria by JTWWW and MD independently. Reasons for exclusion of sources of evidence at full text that do not meet the inclusion criteria will be recorded and reported in the scoping review. Any disagreements that arise between the reviewers at each stage of the selection process will be resolved through discussion or with involvement of a third reviewer if necessary (KJLB or RLM). The results of the search and screening process will be reported in full in the final scoping review, and presented in a PRISMA-ScR flow diagram.

Data extraction

Data will be extracted from the articles by JTWW using a data extraction tool developed by the reviewers. The data extracted will include specific details about the participants, concept, context, study methods and key finding relevant to the review questions. A draft extraction form is provided (see Appendix II). The draft data extraction tool will be modified and revised as necessary during the process of extracting data from each included evidence source. If appropriate, authors of sources of information will be contacted to request missing information.

Data analysis and presentation

Data extracted from the identified studies will be analysed by JTWW with the support of MD, KJLB and RLM. The nature and frequency of all relevant methods that include environmental impacts in HEE and HTA will be described. The characteristics of these methods, and arguments in support and opposition to their use will be described. Where analytical data exists, these will not be synthesised as this is beyond the scope of this review. However, all relevant analytical results will be included and described. A narrative summary will be accompanied by relevant figures and tables. The review will use the PRISMA-ScR checklist to guide writing of the study report. We will not appraise the quality or risk of bias of included studies, consistent with recommended methods for completing scoping reviews.(21,22)

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Conflicts of Interest

The authors declare no conflicts of interest.

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APPENDIX I: SEARCH STRATEGY

Database: Scopus

Datebase: EconLit

Database: Environment Complete

- health* W/5 ("technology assessment*" OR “economic eval*” OR cost*)

  AND

- (environment* W/5 (effect* OR impact* OR sustainab* or assess* or outcome* or implication* or consideration*)) OR (carbon W/3 (emission* OR footprint* OR dioxide)) OR "greenhouse gas*" OR “climate change”

Database: Medline via Ovid

Database: Evidence-Based Medicine Reviews via Ovid

#	Search
1	Technology Assessment, Biomedical/
2	(health* adj5 “technology assessment*”).tw.
3	Public Health/ec, lj [Economics, Legislation & Jurisprudence]
4	Models, economic/
5	(health* adj5 “economic eval*”).tw.
6	“Costs and Cost Analysis”/
7	(health* adj5 cost*).tw.
8	1 or 2 or 3 or 4 or 5 or 6 or 7
9	(environment* adj5 effect*).tw.
10	(environment* adj5 impact*).tw.
11	(environment* adj5 sustainab*).tw.
12	(environment* adj5 assess*).tw.
13	(environment* adj5 outcome*).tw.
14	(environment* adj5 implication*).tw.
15	(environment* adj5 consideration*).tw.
16	(carbon adj3 emission*).tw.
17	Carbon Footprint/
18	(carbon adj3 footprint*).tw.
19	Carbon Dioxide/an [Analysis]
20	(carbon adj3 dioxide).tw.
21	Greenhouse Gases/
22	Greenhouse Effect/
23	“greenhouse gas*”.tw.
24	Climate Change/
25	Global Warming/
26	“climate change”.tw.
27	9 or 10 or 11 or 12 or 13 or 14 or 15 or 16 or 17 or 18 or 19 or 20 or 21 or 22 or 23 or 24 or 25 or 26
28	8 and 27

Appendix II: DATA EXTRACTION TEMPLATE

General Information

- Title
- Lead author
- Corresponding author
- Corresponding author email address
- Country
- Journal/publisher
- Year published
- DOI

Study Details

- Aim(s) of study
- How many methods to include environmental impacts are described? (1, 2, 3, …)
- Is a preferred method stated? (Y/N)
    - If Y, what is the preferred method?

Methodology (<u>repeated per described</u>)

- Method or approach name(s)
- Method description(s)
- Method demonstrated (i.e. is an example given or results presented) (Y/N)
    - Study population
    - Sample size
    - Health condition
    - Technology / intervention
    - Environmental impacts included
    - Unit of value for environmental impacts
    - Other costs included
    - Unit of value for other costs
- Stated benefits of method
- Stated limitations of method
- Stated gaps in knowledge